

Storage Server - Windows Server 2022 : iSCSI

1. Install iSCSI Target
2. Configure iSCSI Target
3. Configure iSCSI Initiator

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Một bộ lưu trữ qua mạng được gọi là iSCSI Target, một Máy Client kết nối với iSCSI Target được gọi là iSCSI Initiator.

iSCSI : Install iSCSI Target

- [1] On CUI installation, Run PowerShell with Admin Privilege and Install iSCSI Target Server.

Windows PowerShell

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install iSCSI Target with admin tools

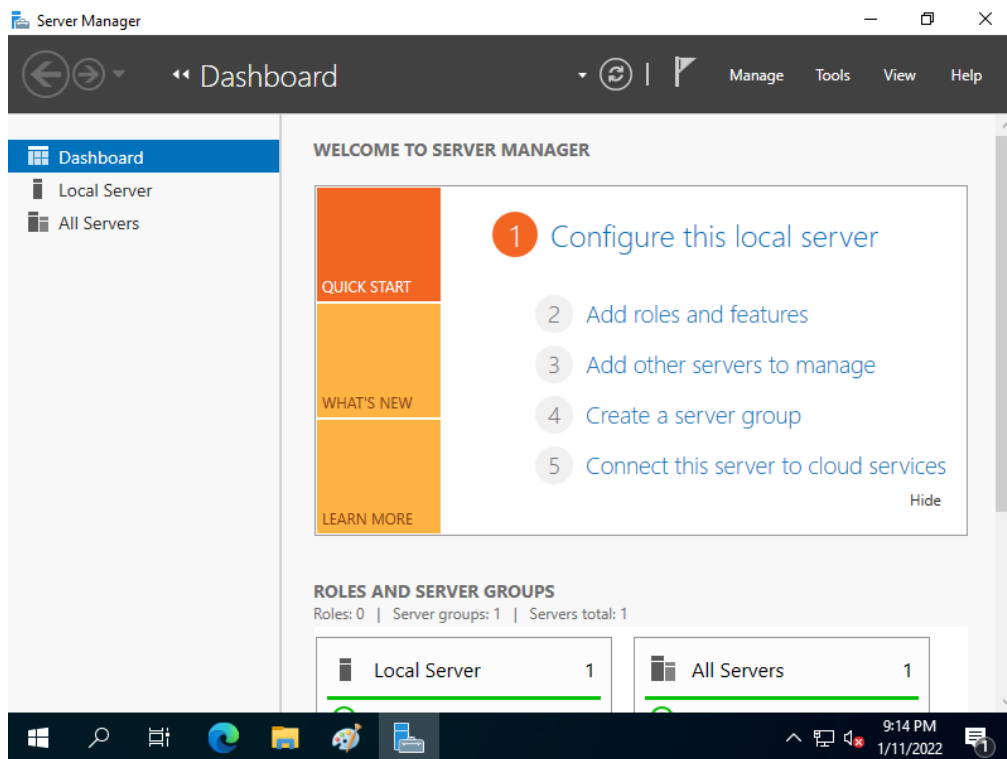
PS C:\Users\Administrator> **Install-WindowsFeature FS-iSCSI-Target-Server -IncludeManagementTools**

Success	Restart Needed	Exit Code	Feature Result
True	No	Success	{File and iSCSI Services, File Server, iSC...

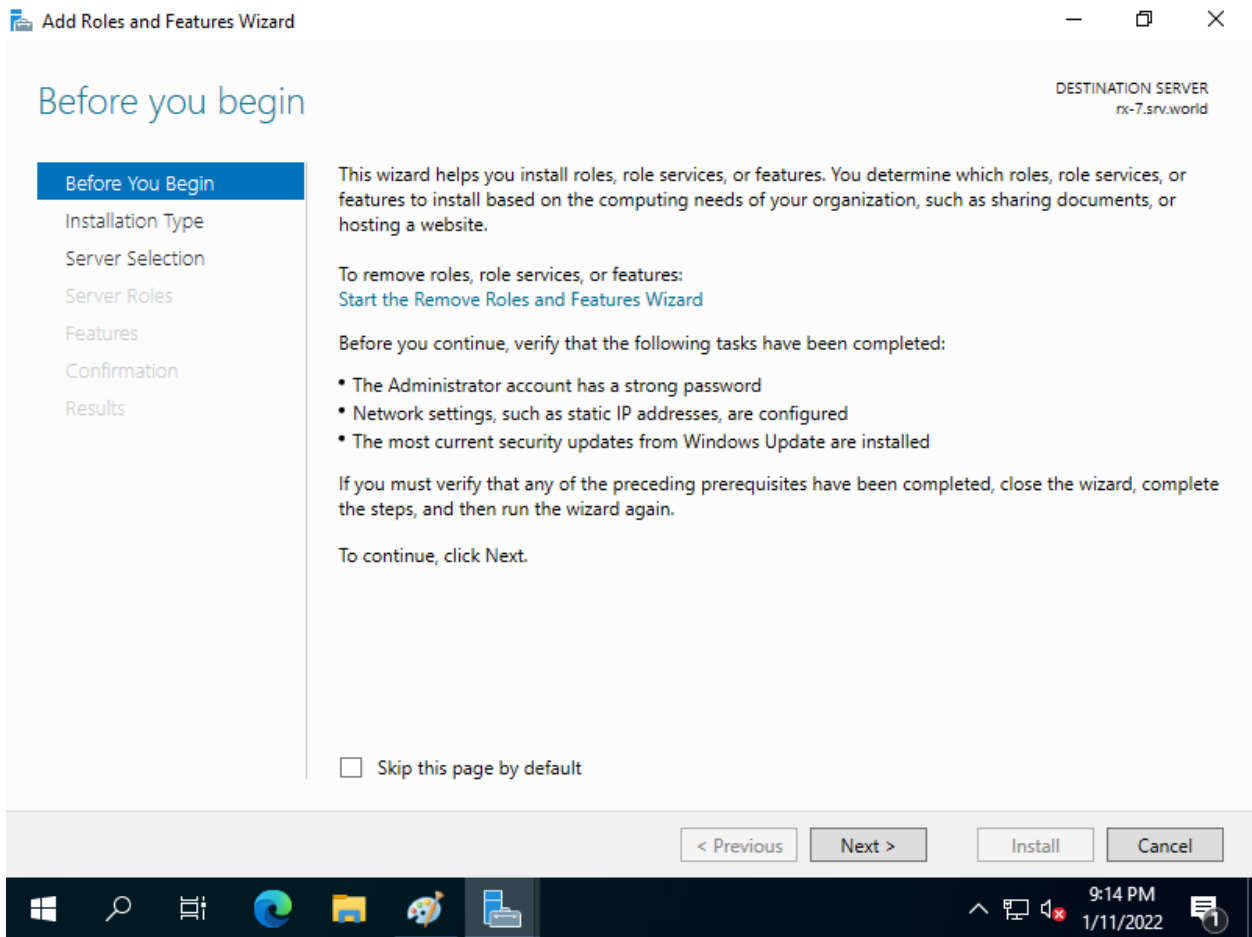
iSCSI : Install iSCSI Target (GUI)

On GUI installation, Set like follows.

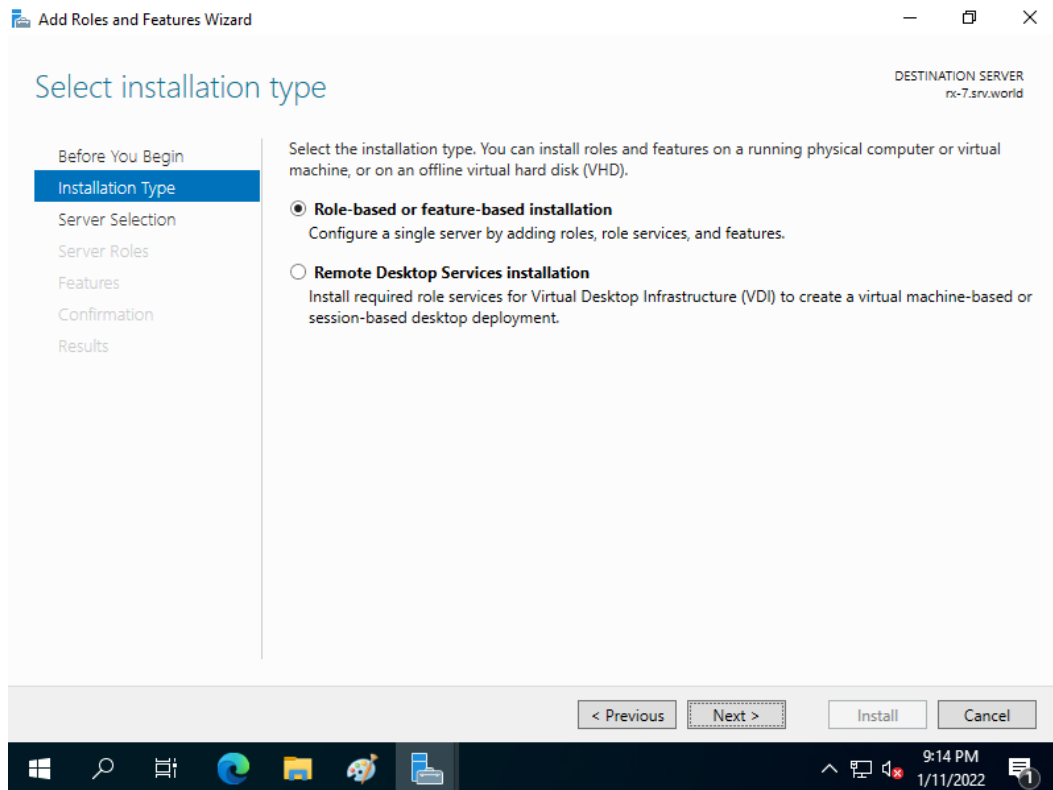
- [2] Run Server Manager and Click [Add roles and features].



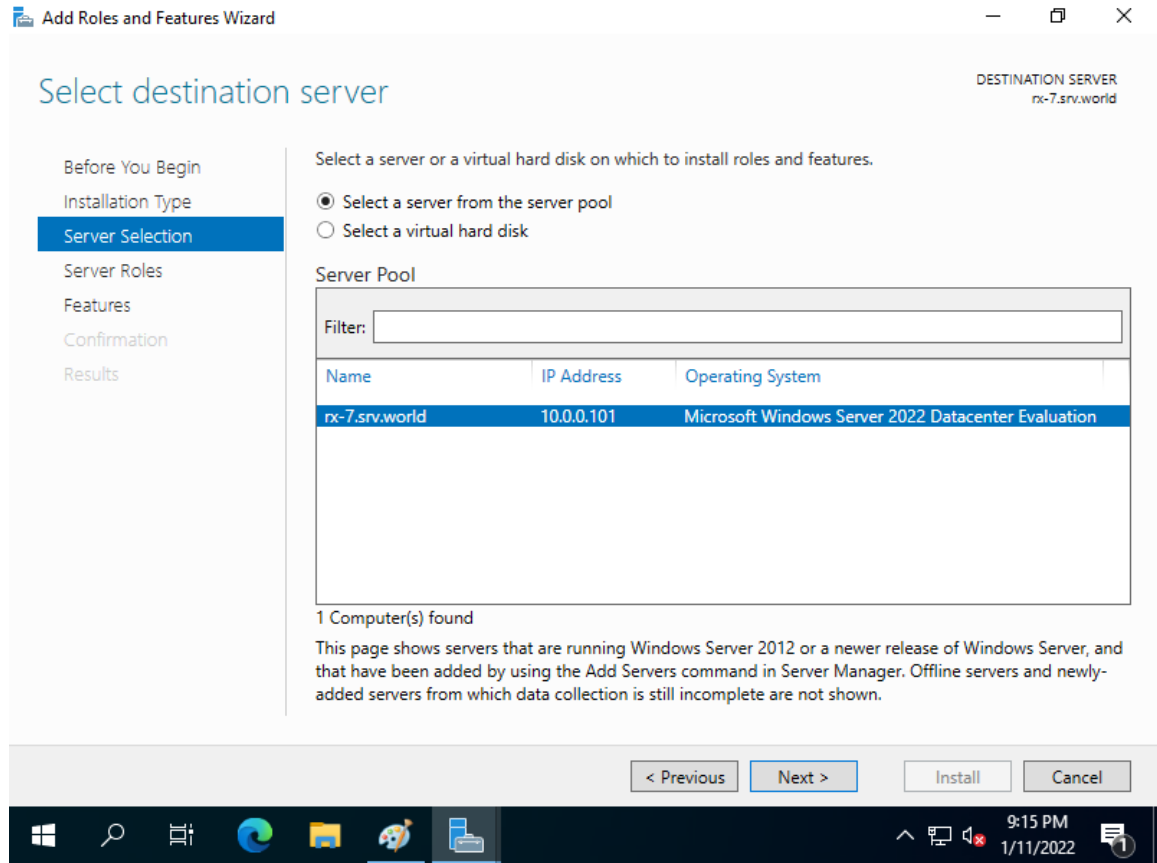
- [3] Click [Next] button.



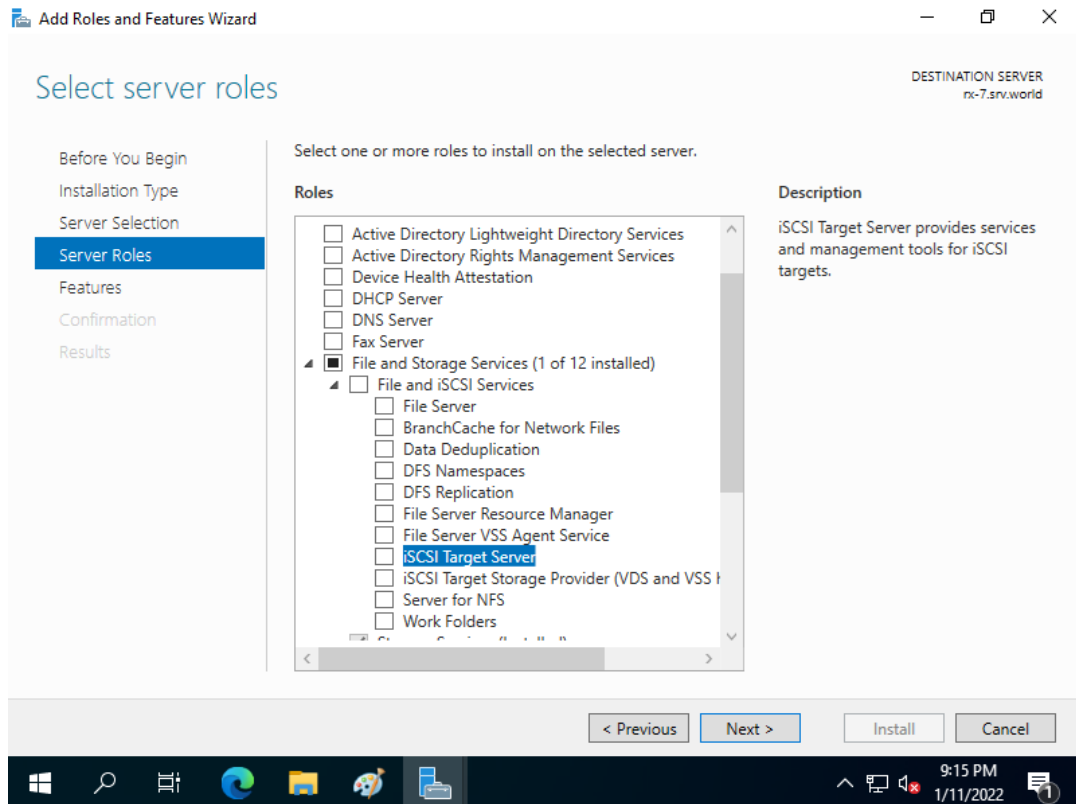
[4] Select [Role-based or feature-based installation].



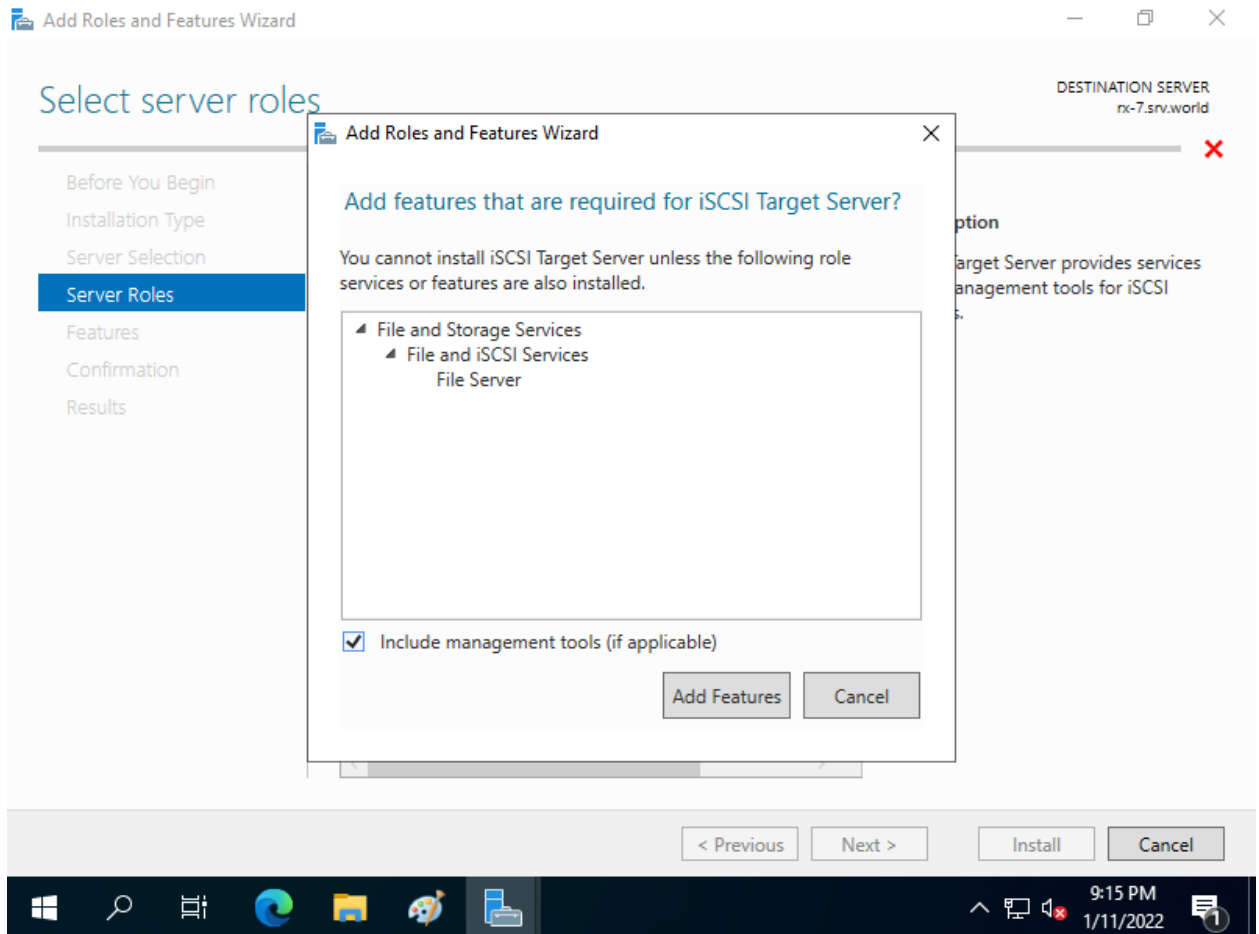
[5] Select a Host which you'd like to add services.



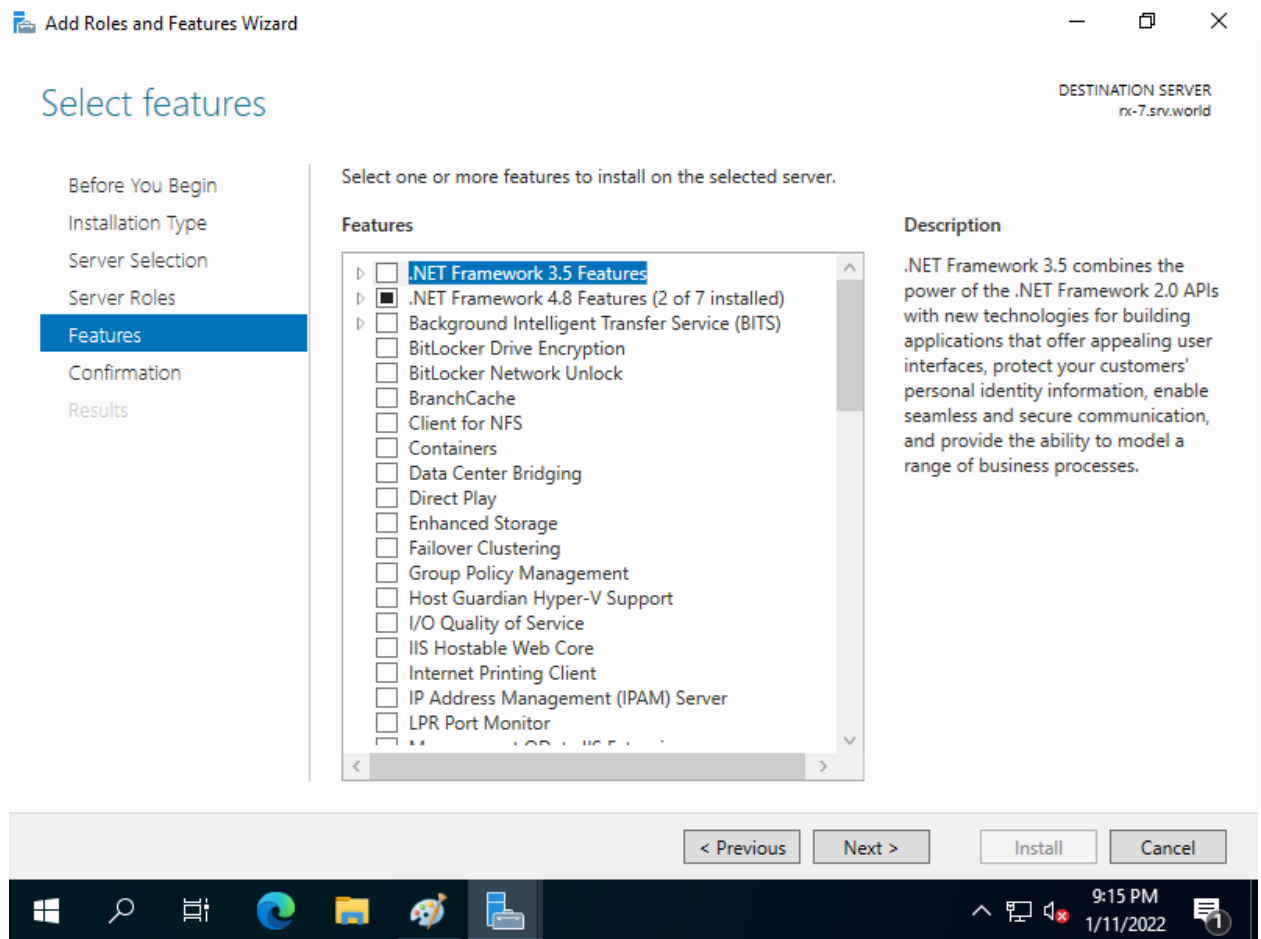
[6] Check a box [iSCSI Target Server].



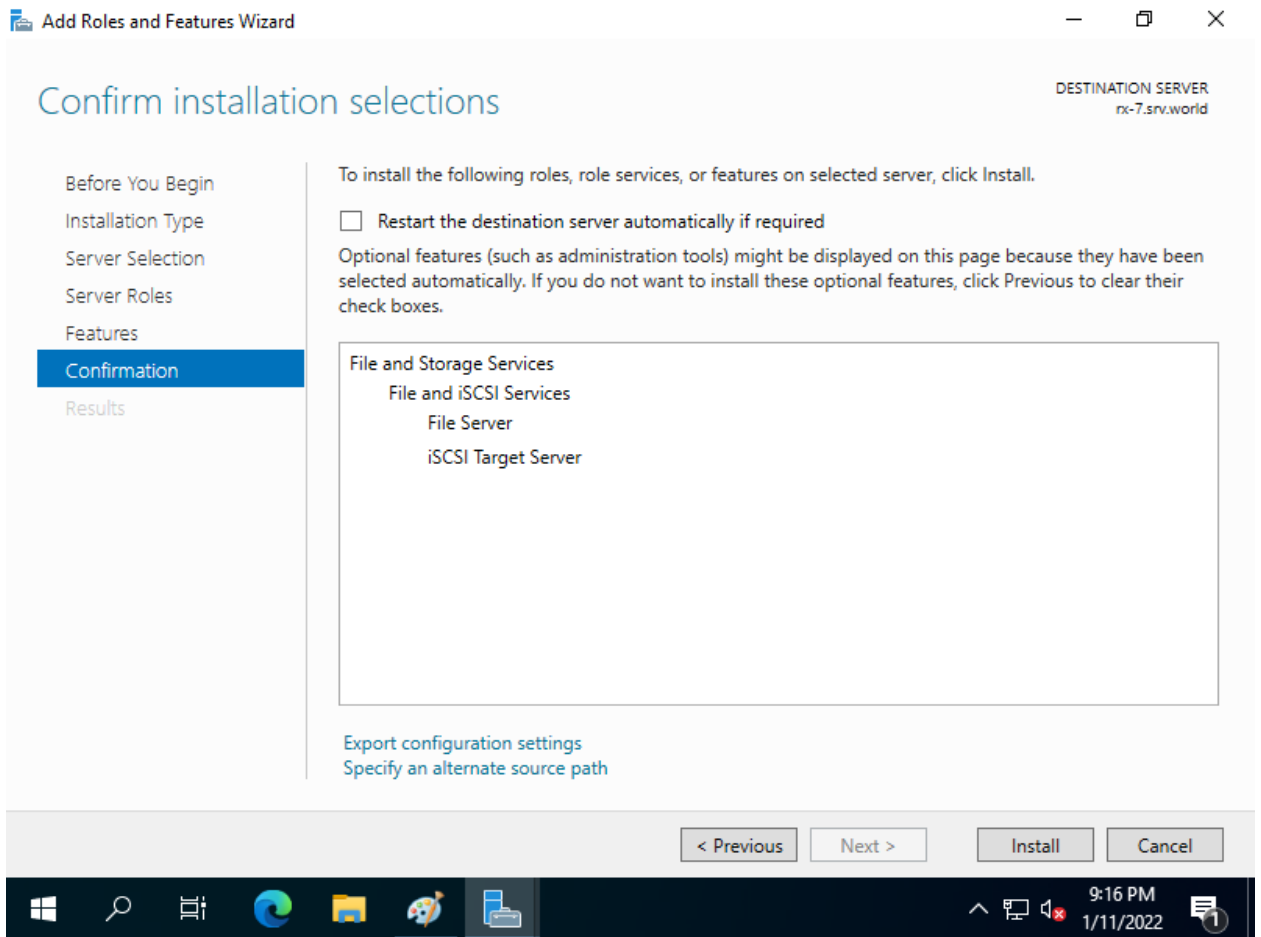
- [7] Additional features are required to add iSCSI Target. Click [Add Features] button and proceed Next.



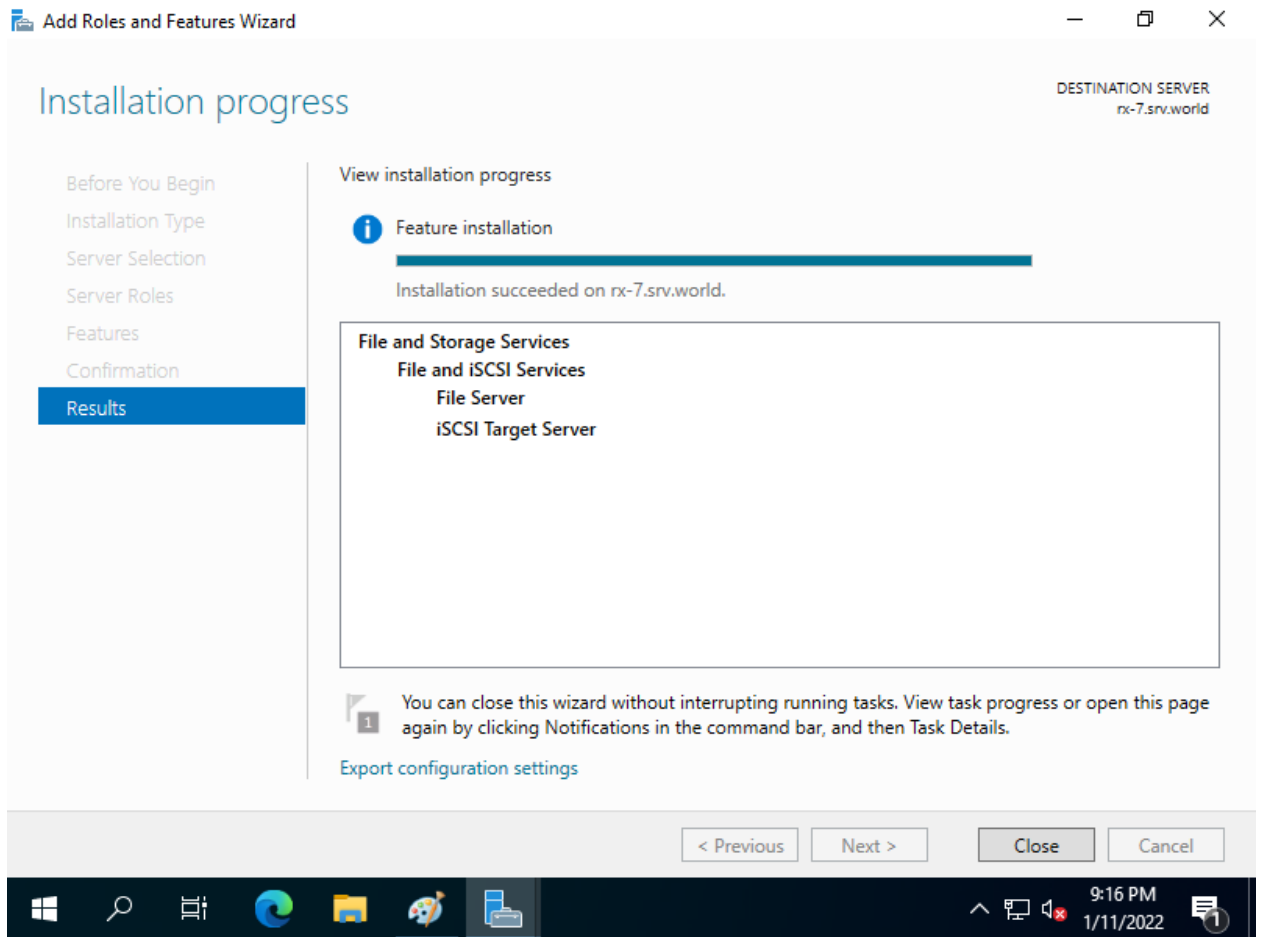
- [8] Click [Next] button.



[9] Click [Install] button.



[10] After finishing installation, click [Close] button.



iSCSI : Configure iSCSI Target

- [1] On CUI Configuration, Run PowerShell with Admin Privilege and Configure iSCSI Target Server.

Windows PowerShell

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create iSCSI Target

[-TargetName] : any Target name

[InitiatorId] : specify Initiators to allow to connect to Target

for specifying Initiators, possible to specify methods like follows not only IPAddress

⇒ IPAddress, IPv6Address, DNSName, IQN, MACAddress

PS C:\Users\Administrator> New-IscsiServerTarget -TargetName "iSCSITarget01" -InitiatorId @("IPAddress:10.0.0.110", "IPAddress:10.0.0.111")

```
ChapUserName      :
ClusterGroupName  :
ComputerName      : rx-7.srv.world
Description       :
EnableChap        : False
EnableReverseChap : False
EnforceIdleTimeoutDetection : True
FirstBurstLength  : 65536
IdleDuration      : 00:00:00
InitiatorIds      : {IPAddress:10.0.0.110, IPAddress:10.0.0.111}
LastLogin         :
LunMappings       : {}
MaxBurstLength    : 262144
MaxReceiveDataSegmentLength : 65536
ReceiveBufferCount : 10
ReverseChapUserName :
Sessions         : {}
Status           : NotConnected
TargetIqn         : iqn.1991-05.com.microsoft:rx-7-iscsitarget01-target
TargetName        : iSCSITarget01
```

create iSCSI virtual disk

[-Path] : any PATH you'd like to create v-disk

[-SizeBytes] : disk size

PS C:\Users\Administrator> New-IscsiVirtualDisk -Path "C:\iSCSIDisk\Disk01.vhdx" -SizeBytes 10GB

```
ClusterGroupName :
ComputerName     : rx-7.srv.world
Description      :
DiskType         : Dynamic
HostVolumId      : {7B618CF5-8E74-4ED2-858A-78A97743E07D}
LocalMountDeviceId :
OriginalPath     :
```

```

ParentPath      :
Path            : C:\iSCSIDisk\Disk01.vhdx
SerialNumber    : B4038C72-8C4A-4F4F-9982-966BB22AE41A
Size           : 10737418240
SnapshotIds     :
Status          : NotConnected
VirtualDiskIndex : 381244894

# assign iSCSI virtual disk to iSCSI Target
PS C:\Users\Administrator> Add-IscsiVirtualDiskTargetMapping -TargetName "iSCSITarget01" -Path "C:\iSCSIDisk\
Disk01.vhdx"

# enable CHAP for iSCSI Target and set Username and Password for authentication
# replace "username", "UserP@ssw0rd01" below to your username and password
# required password length over 12 chars
PS C:\Users\Administrator> Set-IscsiServerTarget `
-TargetName "iSCSITarget01" `
-EnableChap $True `
-Chap (New-Object PSCredential("username", (ConvertTo-SecureString -AsPlainText "UserP@ssw0rd01" -Force))) `
-PassThru

# confirm settings
PS C:\Users\Administrator> Get-IscsiServerTarget -TargetName "iSCSITarget01"

ChapUserName      : username
ClusterGroupName  :
ComputerName      : rx-7.srv.world
Description        :
EnableChap         : True
EnableReverseChap  : False
EnforceIdleTimeoutDetection : True
FirstBurstLength   : 65536
IdleDuration       : 00:05:07
InitiatorIds       : {IPAddress:10.0.0.110, IPAddress:10.0.0.111}
LastLogin          :
LunMappings        : {TargetName:iSCSITarget01;VHD:"C:\iSCSIDisk\Disk01.vhdx";LUN:0}
MaxBurstLength     : 262144
MaxReceiveDataSegmentLength : 65536
ReceiveBufferCount : 10
ReverseChapUserName :
Sessions           : {}
Status             : NotConnected
TargetIqn          : iqn.1991-05.com.microsoft:rx-7-iscsitarget01-target
TargetName         : iSCSITarget01

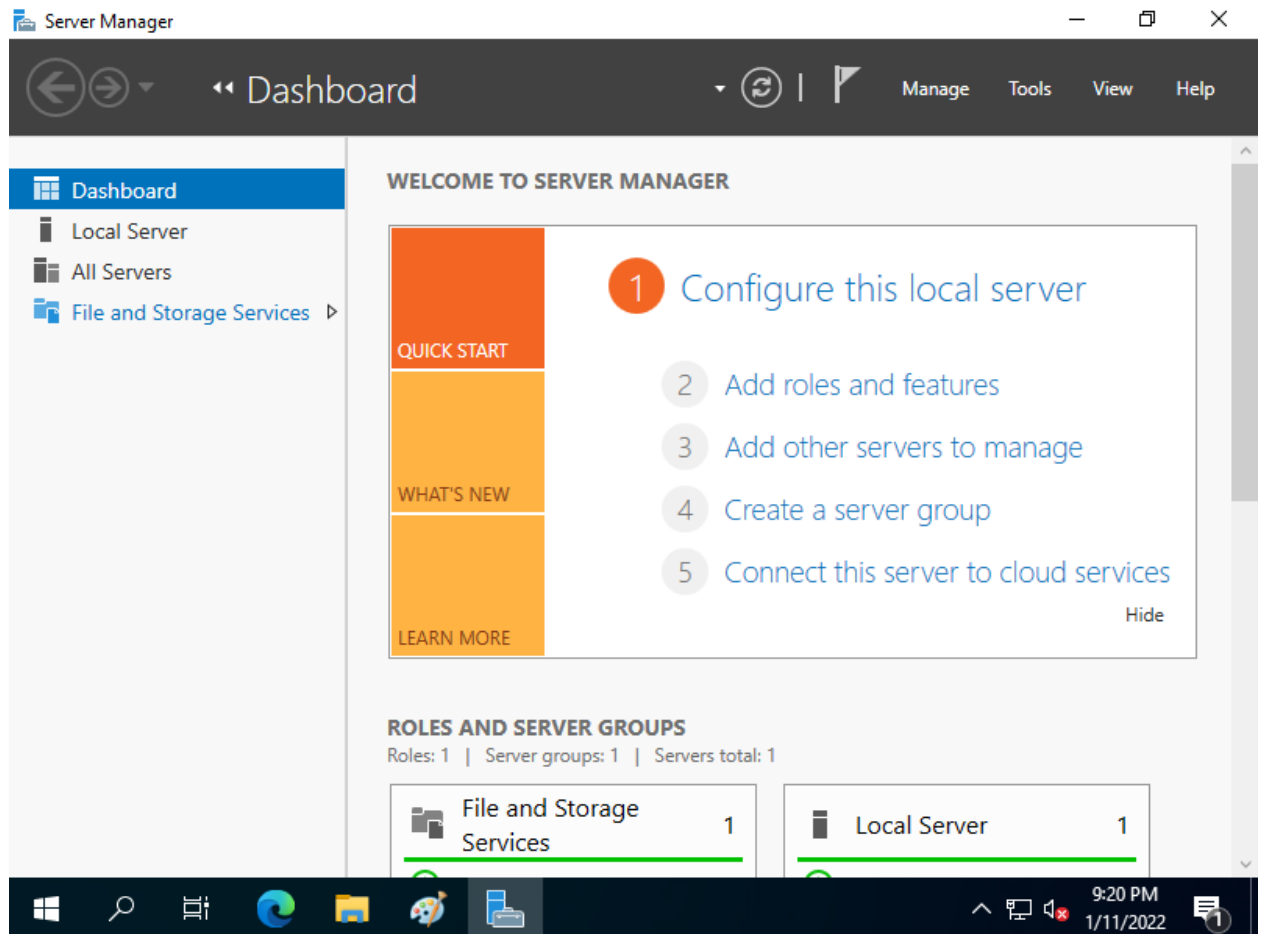
# restart service
PS C:\Users\Administrator> Restart-Service -Name WinTarget

```

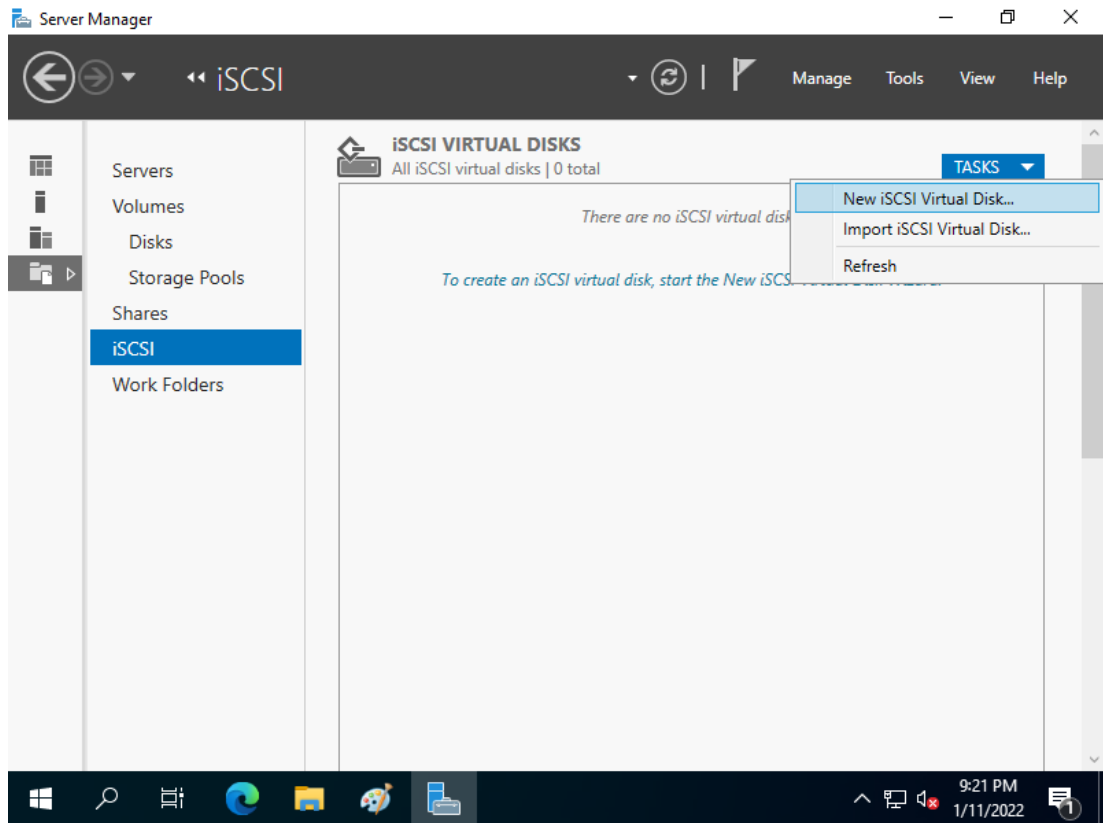
iSCSI : Configure iSCSI Target (GUI)

On GUI configuration, set like follows.

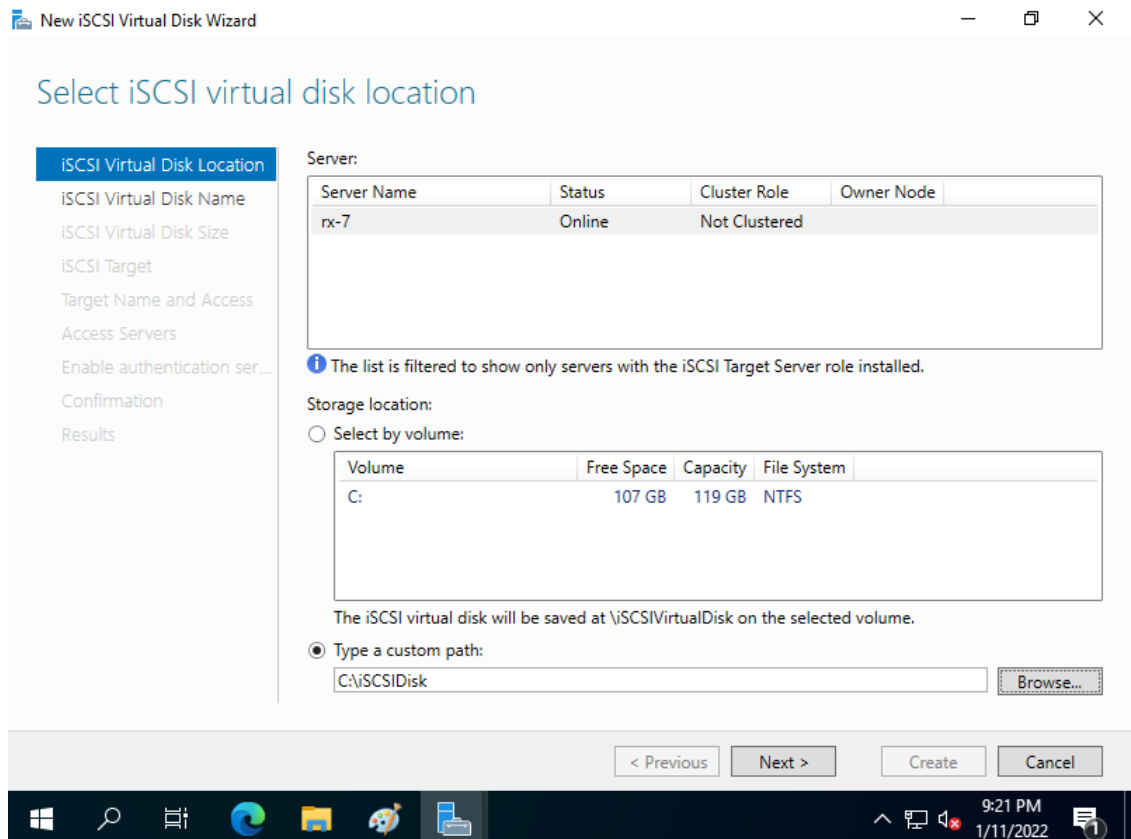
- [2] Run Server Manager and Click [File and Storage Services] on the left pane.



- [3] Select [iSCSI] on the left pane and then click [TASKS] - [New iSCSI Virtual Disk] on the right pane.



- [4] Specify the server and volume for setting iSCSI disk. This example shows to configure a disk on a custom path (C:\iSCSIDisk).



[5] Specify any name for virtual disk.

New iSCSI Virtual Disk Wizard

Specify iSCSI virtual disk name

iSCSI Virtual Disk Location

iSCSI Virtual Disk Name

iSCSI Virtual Disk Size

iSCSI Target

Target Name and Access

Access Servers

Enable authentication ser...

Confirmation

Results

Name:

Description:

Path: C:\SCSIDisk\Disk01.vhdx

< Previous Next > Create Cancel

[6] Specify the disk size for virtual disk. For the selection of the type of disk, choose the one for your requirements.

New iSCSI Virtual Disk Wizard

Specify iSCSI virtual disk size

iSCSI Virtual Disk Location

iSCSI Virtual Disk Name

iSCSI Virtual Disk Size

iSCSI Target

Target Name and Access

Access Servers

Enable authentication ser...

Confirmation

Results

Free space: 107 GB

Size:

☐ Fixed size

This type of disk provides better performance and is recommended for servers running applications with a high level of disk activity. The virtual hard disk is created using the size of the fixed virtual hard disk. It does not change when data is added or deleted.

☒ Clear the virtual disk on allocation

Note: Un-selecting is NOT RECOMMENDED. Clearing a disk to zero will remove any fragments of data that remained on underlying storage, thus protecting from information leaks.

☒ Dynamically expanding

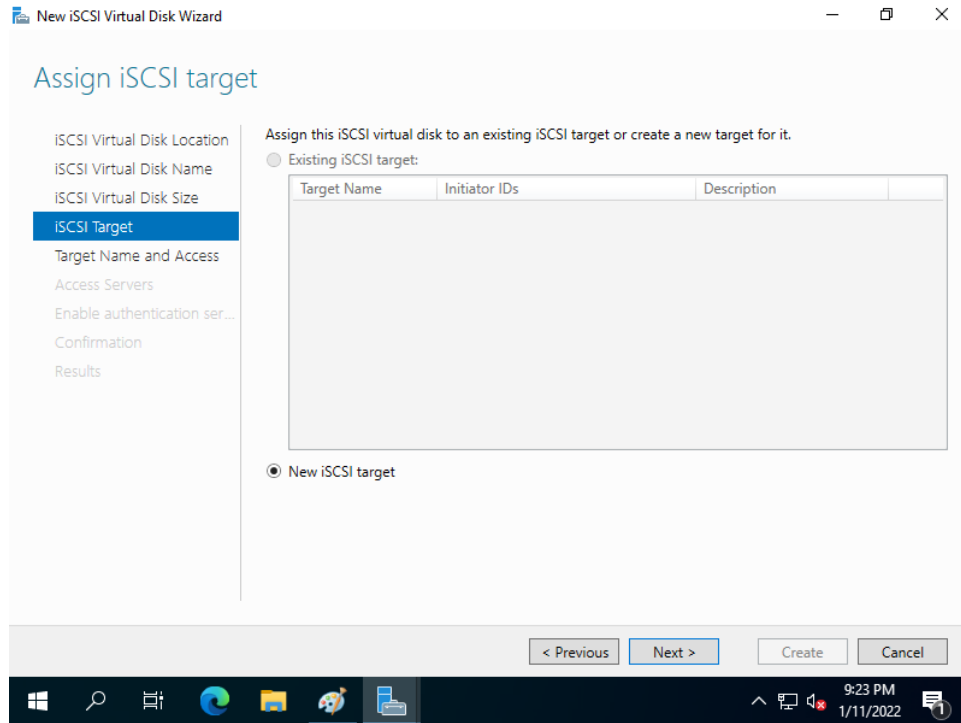
This type of disk provides better use of physical storage space and is recommended for servers running applications that are not disk intensive. The .vhdx file is small when the disk is created and grows as data is written to it.

☐ Differencing

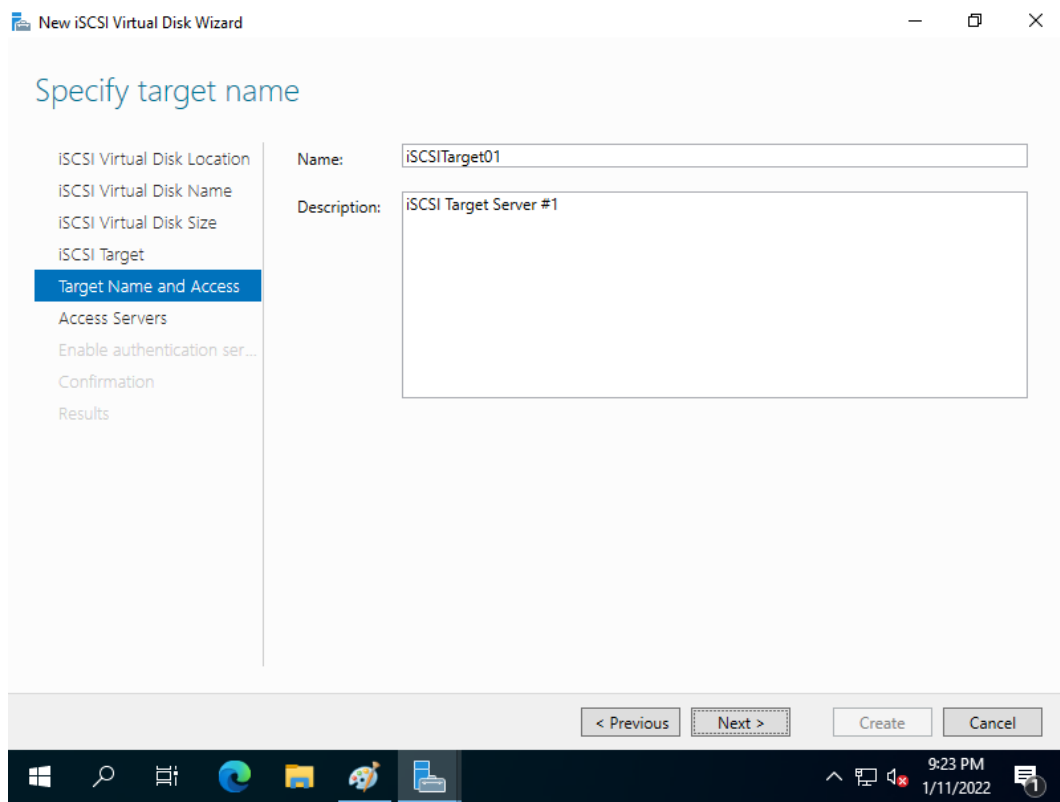
This type of disk is associated in a parent-child relationship with another disk that you want to leave intact. You can make changes to this virtual hard disk without affecting the parent disk and easily revert the changes later.

< Previous Next > Create Cancel

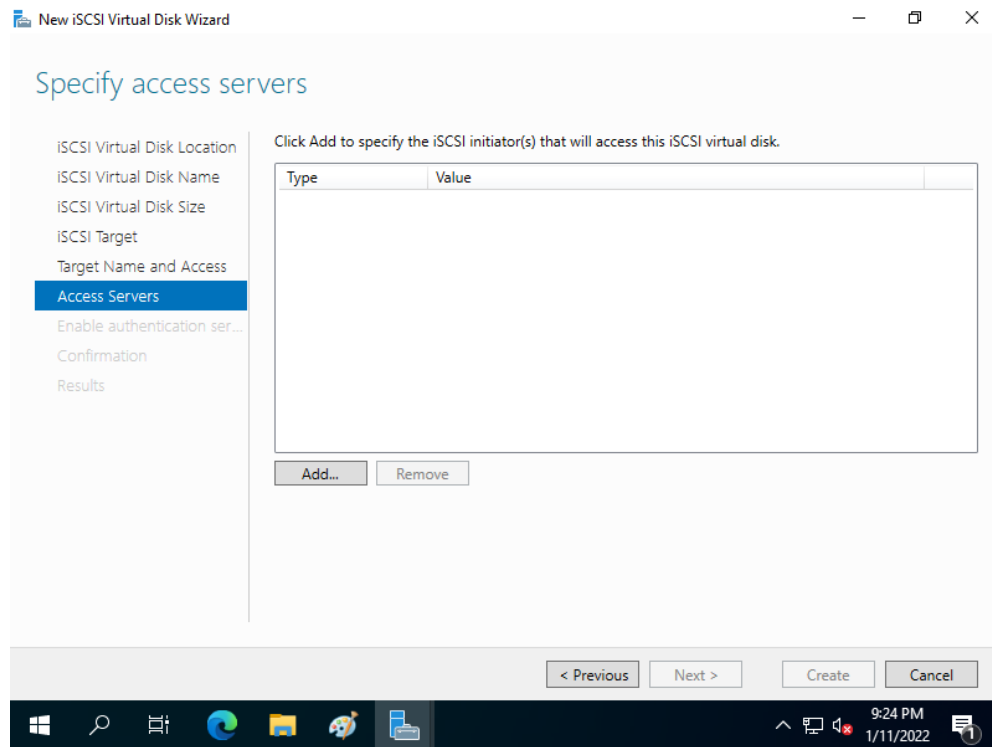
- [7] Specify the iSCSI Target to assign this virtual disk. If the Target is new, select like follows.



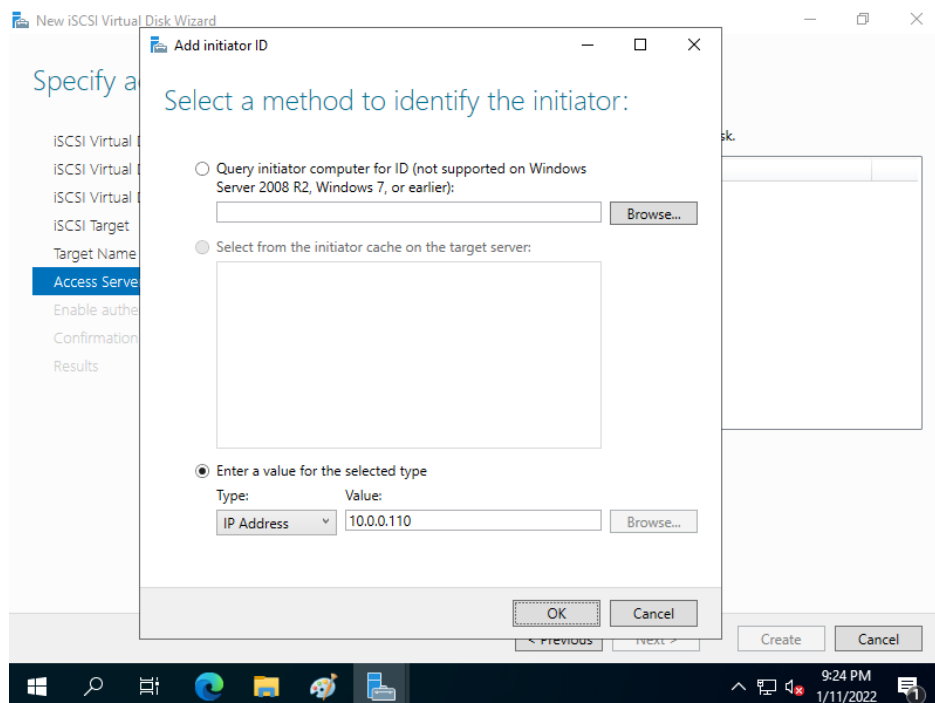
- [8] Specify any name for iSCSI Target.



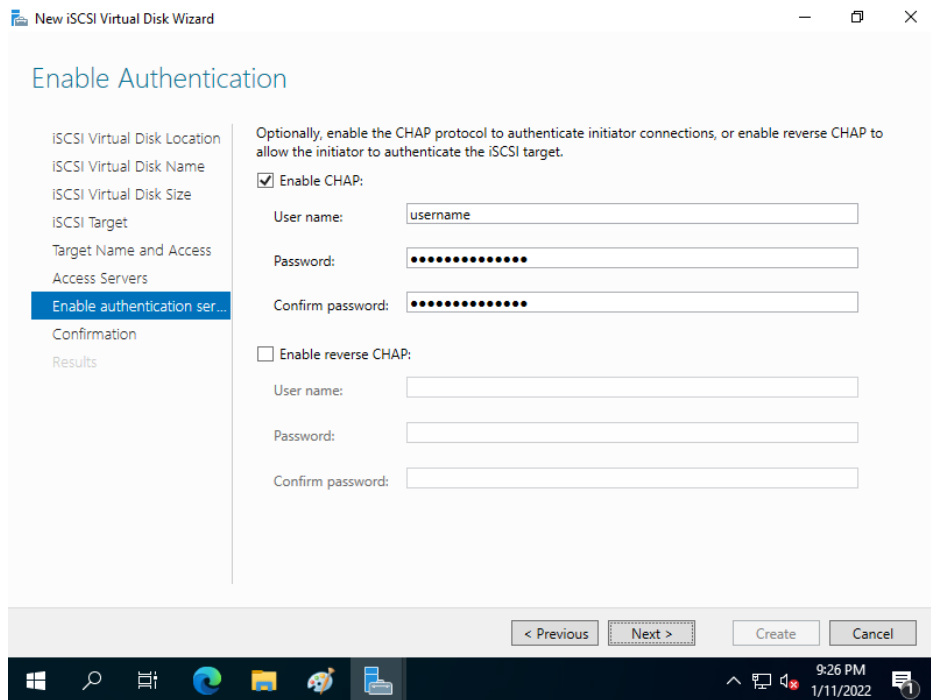
- [9] Specify initiators (clients) to connect to this iSCSI Target. Click [Add...] button.



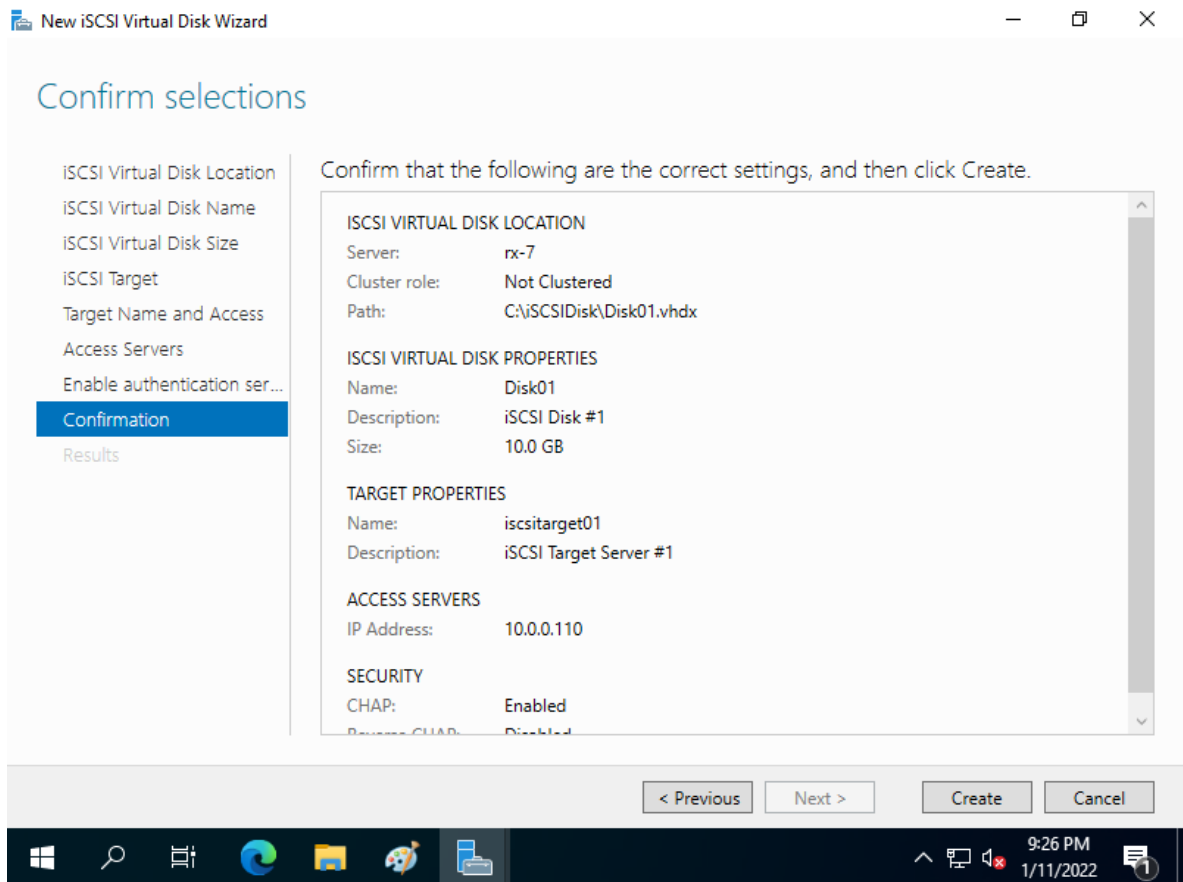
[10] Select a method to identify an initiator. This example shows to set initiators by IP address like follows.



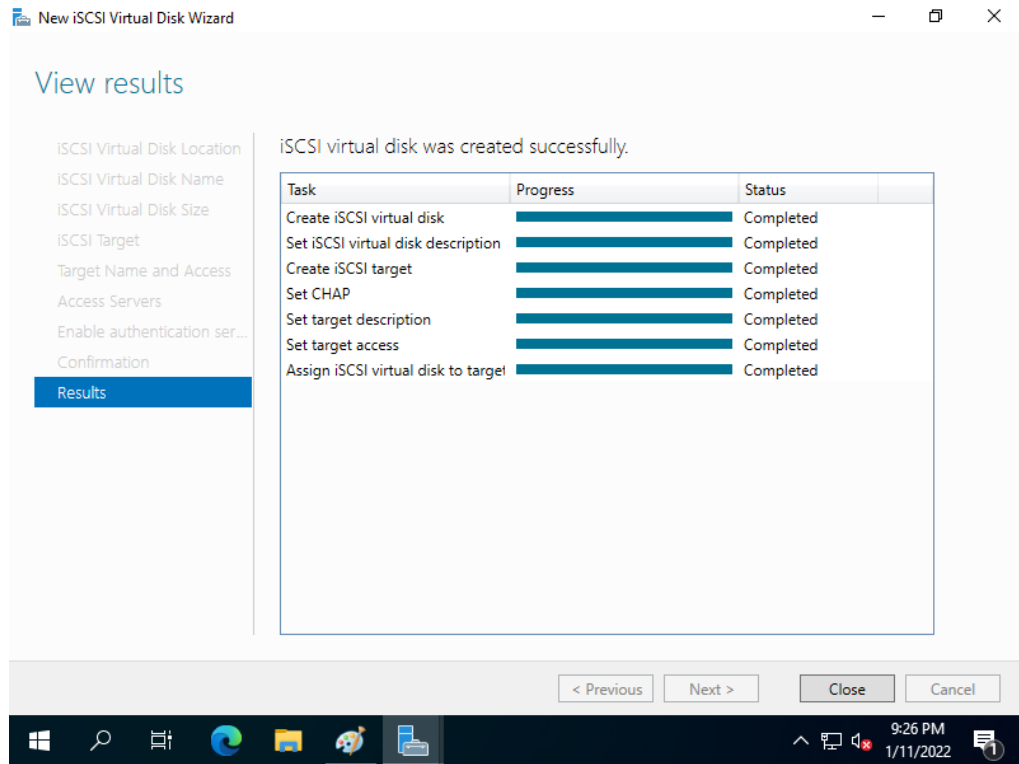
[11] Configure authentication method between Target and Initiator. This example shows to configure CHAP like follows. If select CHAP, set username and password too that are used for connecting to Target from Initiator.



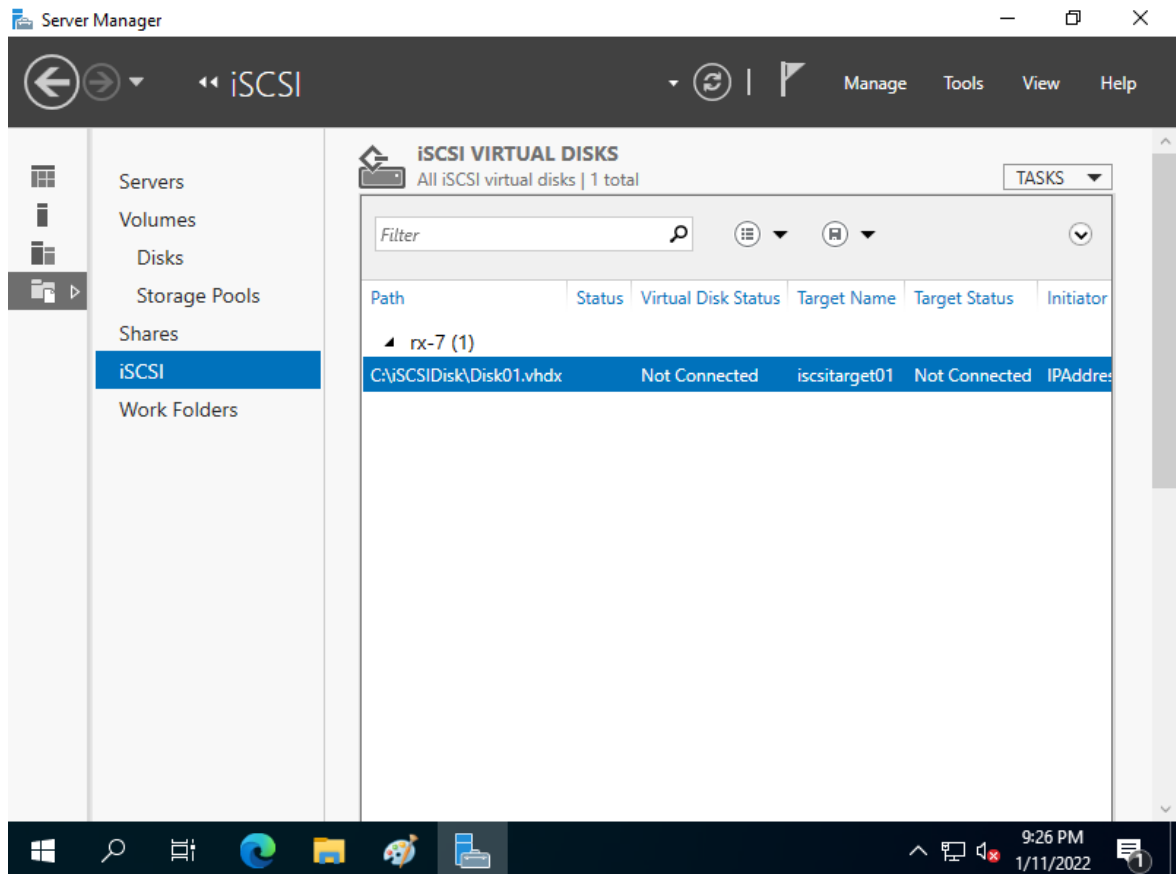
[12] If that's OK all, click [Create] button.



[13] After finishing creation, click [Close] button.



[14] A new virtual disk has been listed like follows.



iSCSI : Configure iSCSI Initiator

Việc thực hiện trên win10/11 cũng tương tự Windows 2022 nhé

- [1] On CUI configuration, Run PowerShell with Admin Privilege and Configure iSCSI Initiator.

```
Windows PowerShell
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# start iSCSI Initiator service & set [automatic] for Startup
PS C:\Users\Administrator> Start-Service -Name MSiSCSI
PS C:\Users\Administrator> Set-Service -Name MSiSCSI -StartupType Automatic

# set iSCSI Target to connect to
PS C:\Users\Administrator> New-IscsiTargetPortal -TargetPortalAddress "10.0.0.101"

InitiatorInstanceName :
InitiatorPortalAddress :
IsDataDigest          : False
IsHeaderDigest         : False
TargetPortalAddress    : 10.0.0.101
TargetPortalPortNumber : 3260
PSComputerName         :

# show iSCSI Target
PS C:\Users\Administrator> Get-IscsiTarget

IsConnected NodeAddress                PSComputerName
-----
False iqn.1991-05.com.microsoft:rx-7-iscsitarget01-target

# connect to iSCSI Target
# [-NodeAddress] : the name confirmed above
# [-ChapUsername] : username that you set on iSCSI Target Config
# [-ChapSecret] : password of the user above
PS C:\Users\Administrator> Connect-IscsiTarget `
-NodeAddress iqn.1991-05.com.microsoft:rx-7-iscsitarget01-target `
-AuthenticationType ONEWAYCHAP `
-ChapUsername "username" `
-ChapSecret "UserP@ssw0rd01" `
-IsPersistent $True

AuthenticationType : ONEWAYCHAP
InitiatorInstanceName : ROOT\ISCSIPRT\0000_0
InitiatorNodeAddress : iqn.1991-05.com.microsoft:rx-9.srv.world
InitiatorPortalAddress : 0.0.0.0
InitiatorSidIdentifier : 400001370000
IsConnected : True
```

```

IsDataDigest      : False
IsDiscovered      : True
IsHeaderDigest    : False
IsPersistent      : True
NumberOfConnections : 1
SessionIdentifier  : ffff8004375fe010-4000013700000002
TargetNodeAddress  : iqn.1991-05.com.microsoft:rx-7-iscsitarget01-target
TargetSidIdentifier : 0100
PSComputerName    :

```

show established connection

```
PS C:\Users\Administrator> Get-IscsiConnection
```

```
ConnectionIdentifier : ffff8004375fe010-1
```

```
InitiatorAddress      : 0.0.0.0
```

```
InitiatorPortNumber   : 19411
```

```
TargetAddress         : 10.0.0.101
```

```
TargetPortNumber      : 3260
```

```
PSComputerName       :
```

show disks

on the example blow, [Number 1] is the iSCSI disk

```
PS C:\Users\Administrator> Get-Disk | Format-Table -AutoSize -Wrap
```

Number	Friendly Name	Serial Number	HealthStatus	OperationalStatus	Total Size	Partition Style
1	MSFT Virtual HD	B4038C72-8C4A-4F4F-9982-966BB22AE41A	Healthy	Offline	10 GB	RAW
0	VMware Virtual NVMe Disk	VMWare NVME_0000	Healthy	Online	120 GB	GPT

[2] To enable to read or write iSCSI disks, set like follows.

turn it online

```
PS C:\Users\Administrator> Set-Disk -Number 1 -IsOffline $False
```

initialize it with GPT partitoon type

```
PS C:\Users\Administrator> Initialize-Disk -Number 1 -PartitionStyle GPT
```

confirm settings

```
PS C:\Users\Administrator> Get-Disk | Format-Table -AutoSize -Wrap
```

Number	Friendly Name	Serial Number	HealthStatus	OperationalStatus	Total Size	Partition Style
1	MSFT Virtual HD	B4038C72-8C4A-4F4F-9982-966BB22AE41A	Healthy	Online	10 GB	GPT
0	VMware Virtual NVMe Disk	VMWare NVME_0000	Healthy	Online	120 GB	GPT

assign drive letter and create partition

```
PS C:\Users\Administrator> New-Partition -DiskNumber 1 -UseMaximumSize -AssignDriveLetter
```

```
DiskPath: \\?\scsi#disk&ven_msft&prod_virtual_hd#1&1c121344&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}
```

PartitionNumber	DriveLetter	Offset	Size	Type
2	E	16777216	9.98 GB	Basic

format with NTFS

```
PS C:\Users\Administrator> Format-Volume -DriveLetter E -FileSystem NTFS -Force
```

DriveLetter	FriendlyName	FileSystemType	DriveType	HealthStatus	OperationalStatus	SizeRemaining	Size
E		NTFS	Fixed	Healthy	OK	9.95 GB	9.98 GB

confirm settings

```
PS C:\Users\Administrator> Get-Volume | Format-Table -AutoSize -Wrap
```

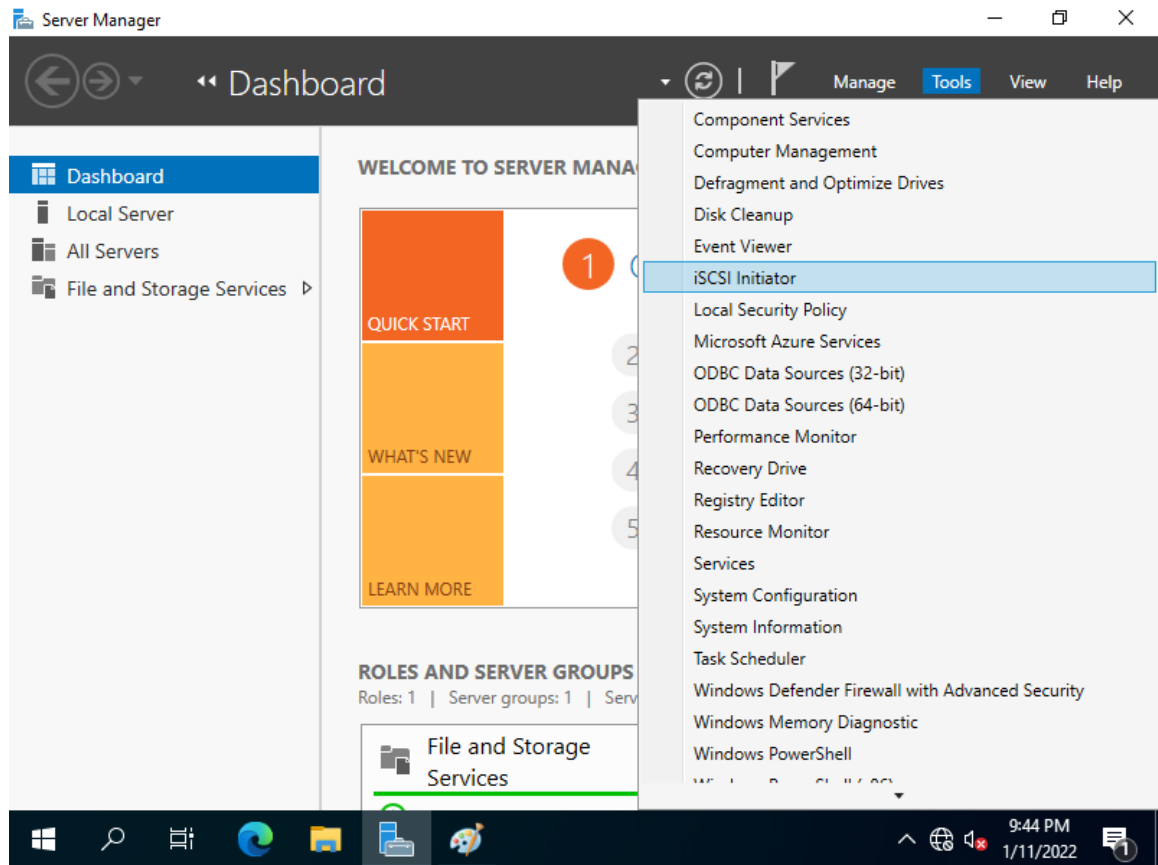
DriveLetter	FriendlyName	FileSystemType	DriveType	HealthStatus	OperationalStatus	SizeRemaining	Size
		NTFS	Fixed	Healthy	OK	85.68 MB	523 MB
		FAT32	Fixed	Healthy	OK	67.3 MB	96 MB
E		NTFS	Fixed	Healthy	OK	9.95 GB	9.98 GB
C		NTFS	Fixed	Healthy	OK	105.42 GB	119.37 GB
D		Unknown	CD-ROM	Healthy	Unknown	0 B	0 B

iSCSI : Configure iSCSI Initiator (GUI)

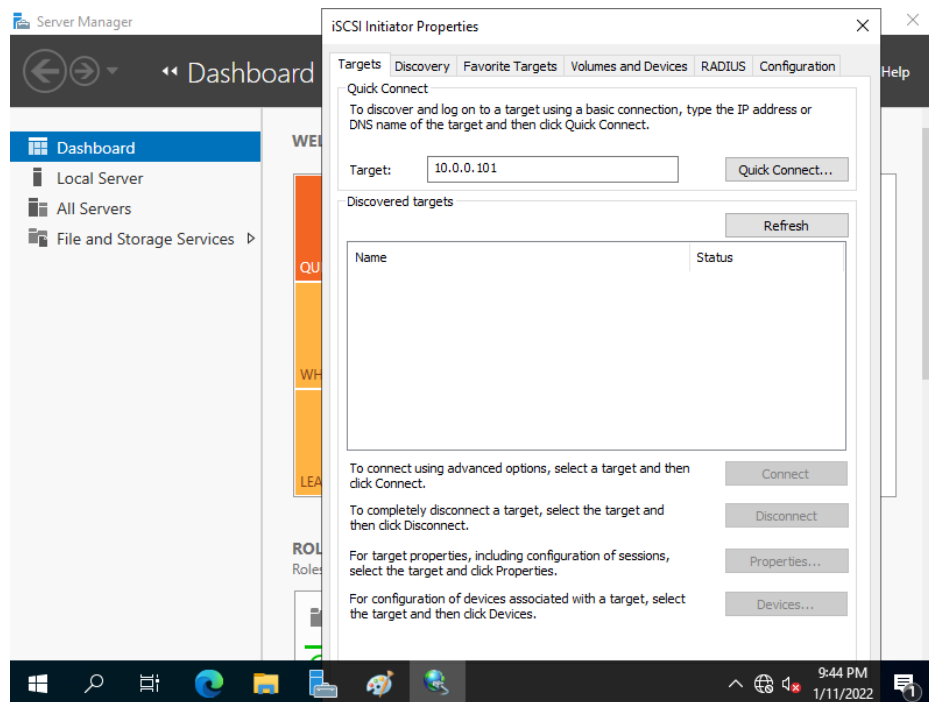
On GUI configuration, set like follows.

- [3] Run Server Manager and Open [Tools] - [iSCSI Initiator].

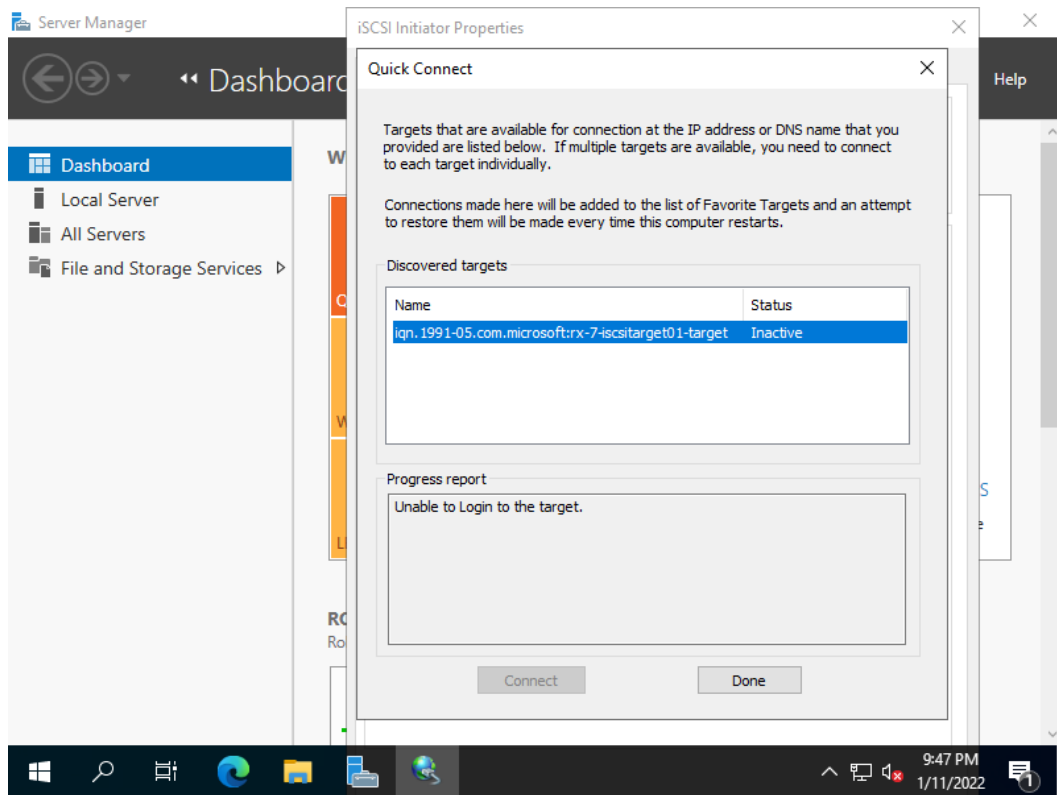
On initial starting, confirmation message is shown because iSCSI Initiator service is not enabled by default.



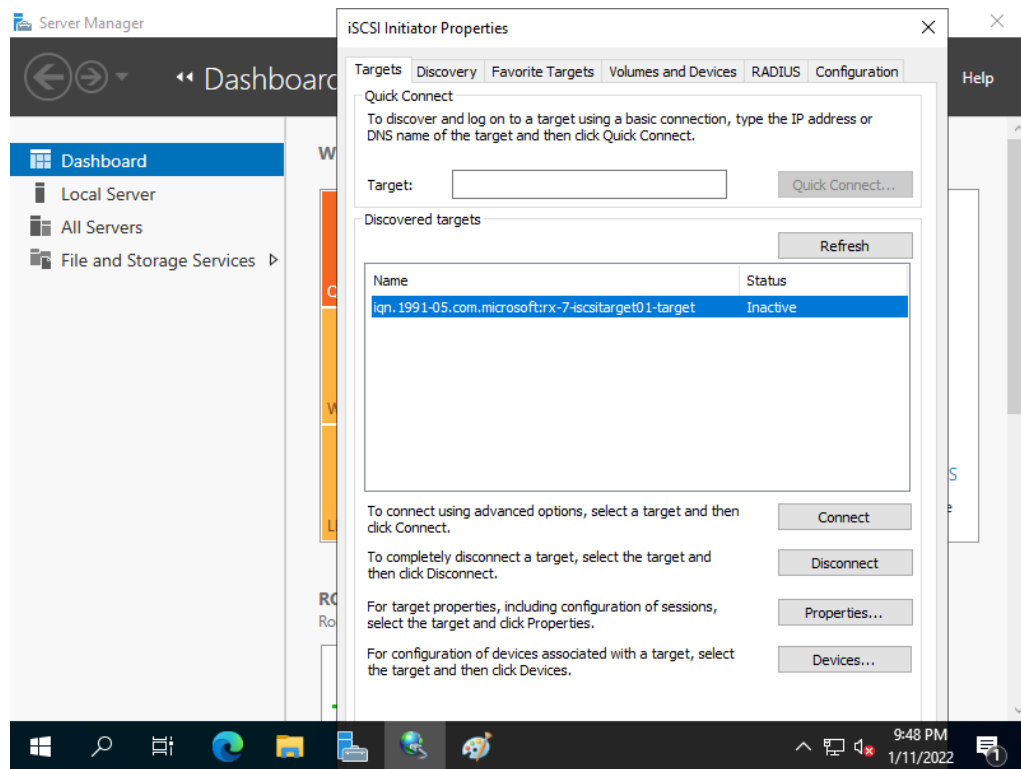
[4] Input Target's Hostname or IP address on [Target] field and click [Quick Connect] button.



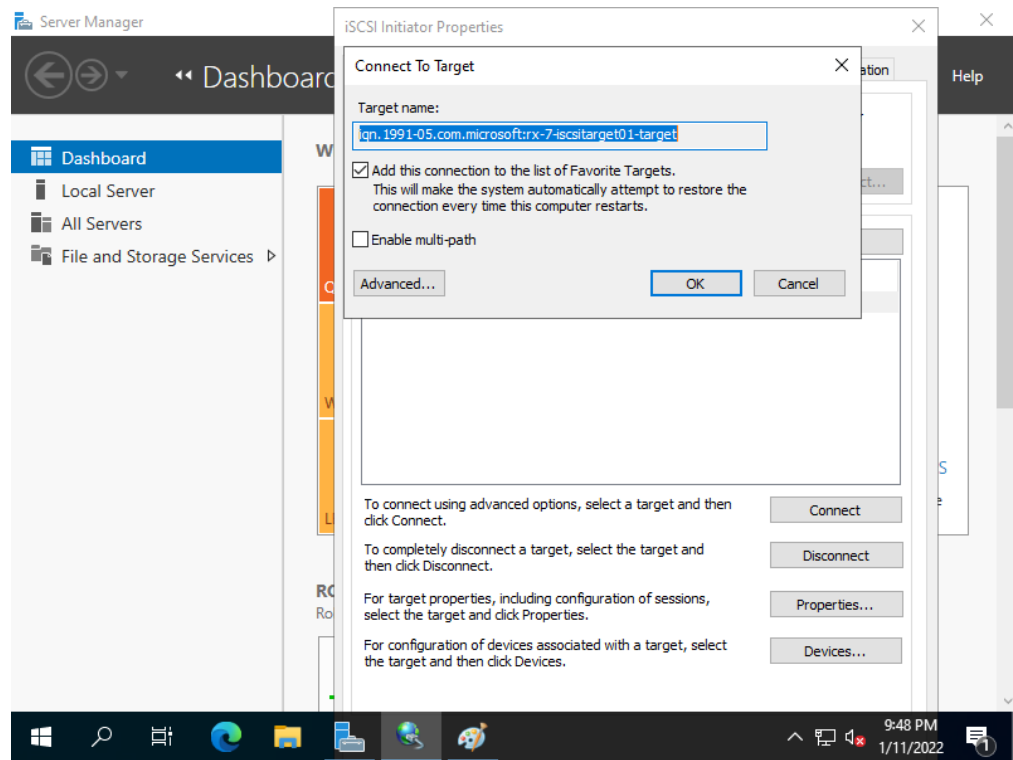
[5] iSCSI Target has been detected like follows, click [Done] button.



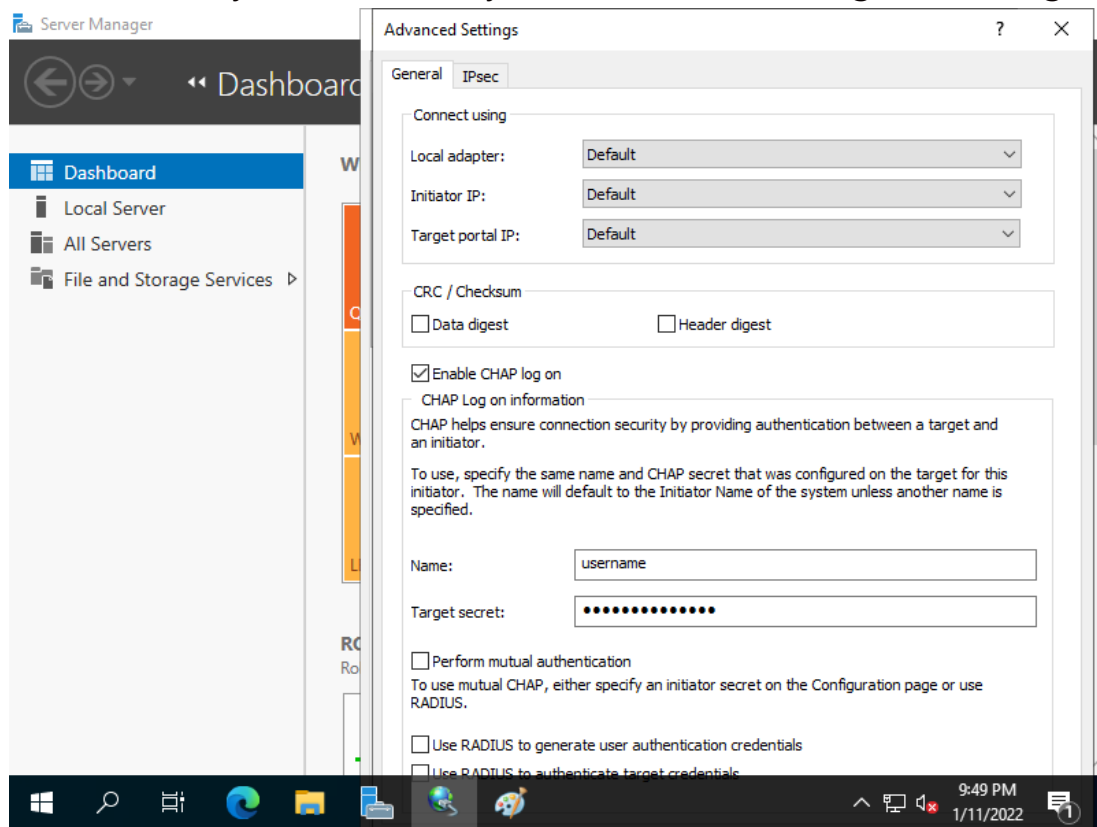
[6] Click [Connect] button.



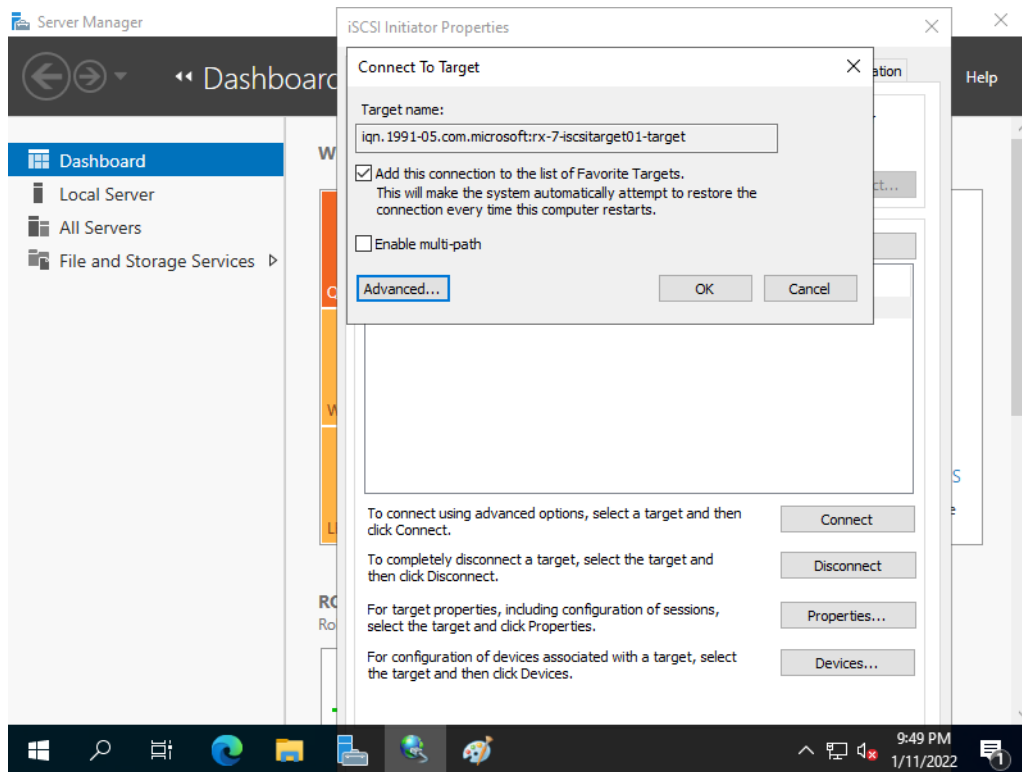
[7] Click [Advanced] button.



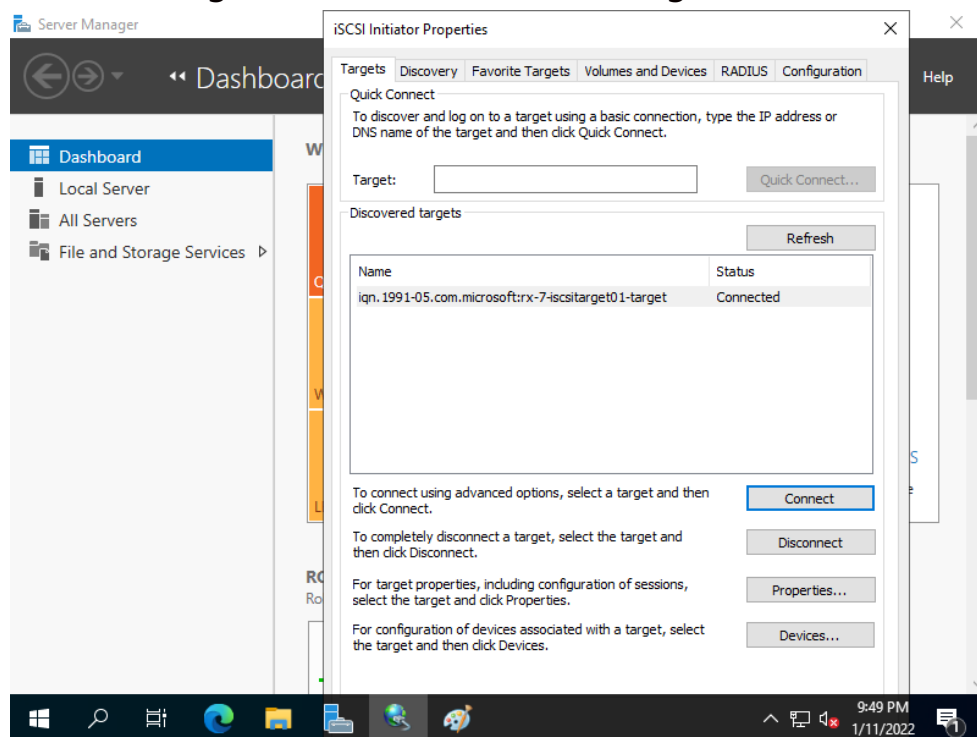
- [8] Check a box [Enable CHAP log on] and specify username and password on the fields. They are the one you set on iSCSI Target's configuration.



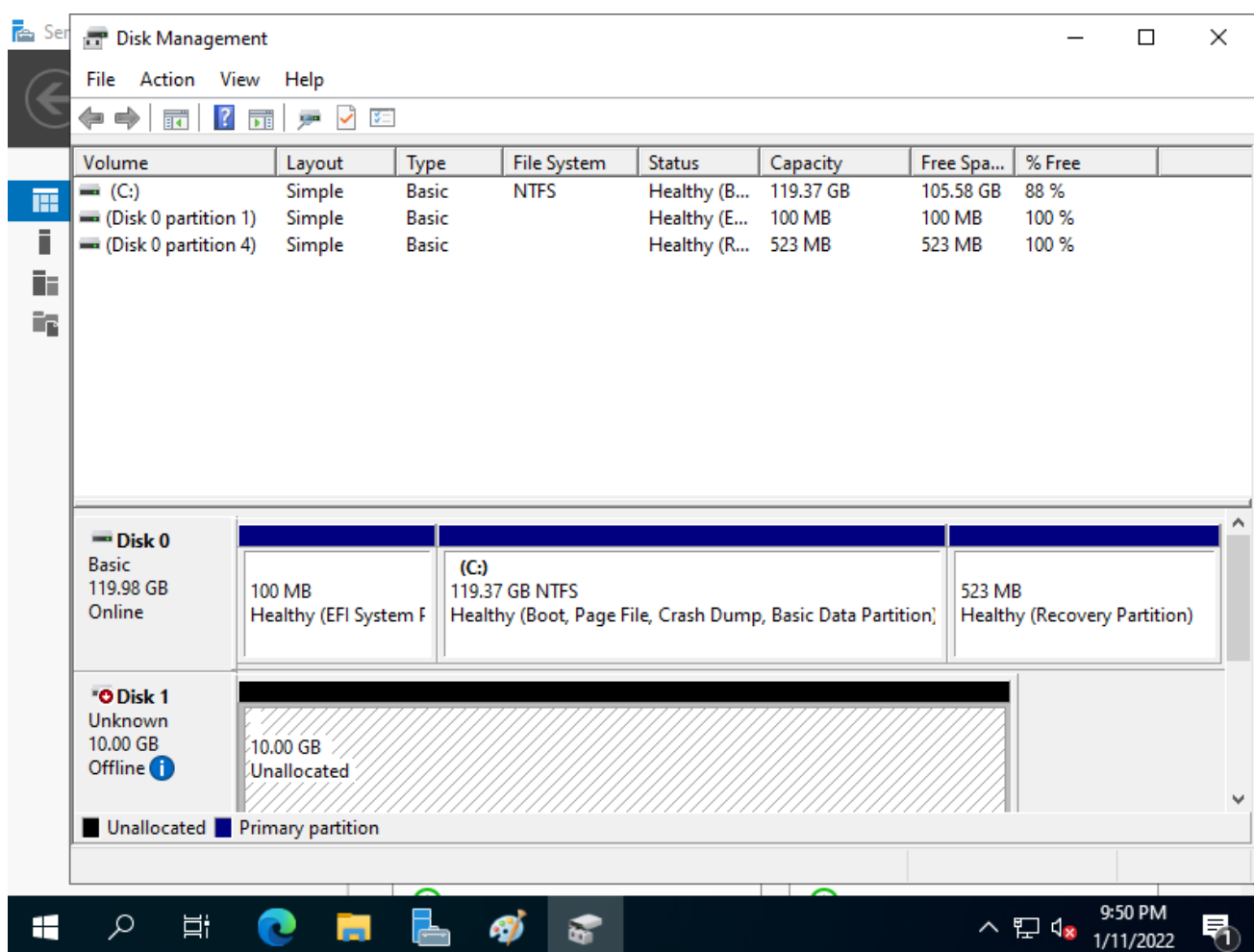
- [9] Click [OK] button.



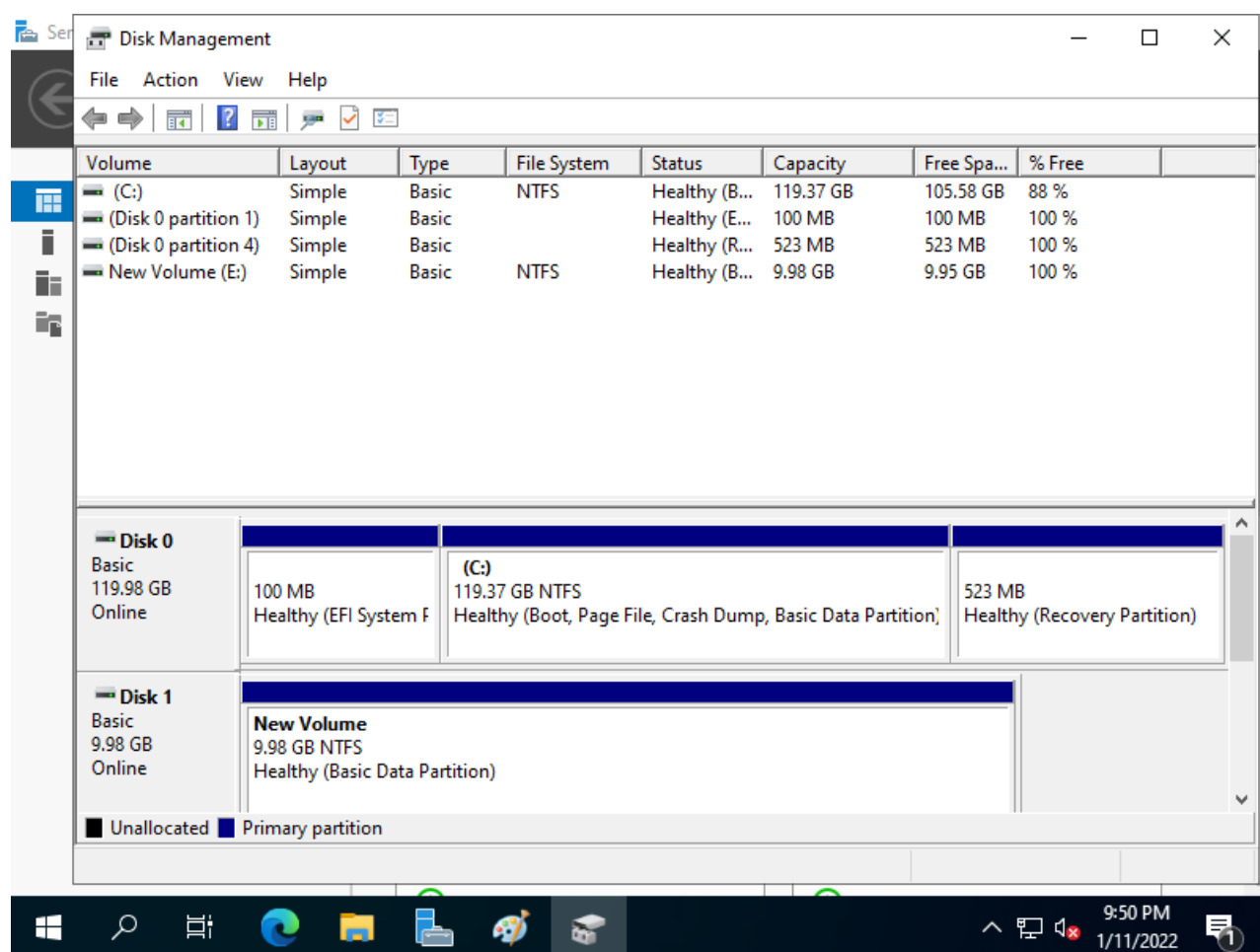
[10] If successfully connected, the status turns to [Connected] like follows. That's OK to configure iSCSI initiator settings.



[11] On the [Disk Management] tool, the iSCSI Disk has been attached like follows.



[12] To use the iSCSI Disk, turn it online and initialize it and then assign drive letter to it.



THE END & CONTACT

Thanks for reading.

If you have any question, please send to me via

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3. Skype: bienbien.it | Viber: 0975277603 | Zalo: 0975277603